Thinking Corner

1. When will the values of $\sin\theta$ and $\cos\theta$ be equal?
2. For what values of θ, $\sin\theta = 2$?
3. Among the six trigonometric quantities, as the value of angle θ increase from $0°$ to $90°$, which of the six trigonometric quantities has undefined values?
4. Is it possible to have eight trigonometric ratios?
5. Let $0° \leq \theta \leq 90°$. For what values of θ does

   (i) $\sin\theta > \cos\theta$ (ii) $\cos\theta > \sin\theta$ (iii) $\sec\theta = 2\tan\theta$ (iv) $\operatorname{cosec}\theta = 2\cot\theta$

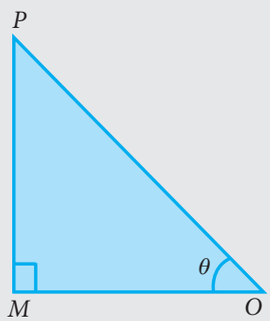
6.2 Trigonometric Identities

For all real values of θ, we have the following three identities.

(i) $\sin^2\theta + \cos^2\theta = 1$ (ii) $1 + \tan^2\theta = \sec^2\theta$ (iii) $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

These identities are termed as three fundamental identities of trigonometry.

We will now prove them as follows.

Picture	Identity	Proof
 Fig. 6.3	$\sin^2\theta + \cos^2\theta = 1$	In the right angled ΔOMP, we have $\frac{OM}{OP} = \cos\theta, \quad \frac{PM}{OP} = \sin\theta$ ...(1) By Pythagoras theorem $MP^2 + OM^2 = OP^2$ ...(2) Dividing each term on both sides of (2) by OP^2, $(\because OP \neq 0)$ we get, $\frac{MP^2}{OP^2} + \frac{OM^2}{OP^2} = \frac{OP^2}{OP^2}$ $\Rightarrow \left(\frac{MP}{OP}\right)^2 + \left(\frac{OM}{OP}\right)^2 = \left(\frac{OP}{OP}\right)^2$ From (1), $(\sin\theta)^2 + (\cos\theta)^2 = 1^2$ Hence $\sin^2\theta + \cos^2\theta = 1$ In the right angled ΔOMP, we have $\frac{MP}{OM} = \tan\theta, \quad \frac{OP}{OM} = \sec\theta$ ...(3)

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